
Context Construction Strategies in Medical RAG: A Systematic Review of Chunking Methods for Biomedical Retrieval-Augmented Generation

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Abstract

Retrieval-Augmented Generation (RAG) has become the dominant architecture for grounding large language model (LLM) outputs in verified external knowledge. Central to every RAG pipeline is the process of chunking — the division of source documents into retrievable units. While a substantial body of work has evaluated chunking in general-domain settings, the biomedical domain presents distinctive challenges: dense entity co-occurrence, abbreviation-rich clinical prose, and the structural peculiarities of PubMed abstracts. This paper reviews the state of the art in chunking strategies for RAG, with particular attention to the gap between general-domain findings and biomedical applicability. We survey fixed-size, sentence-boundary, semantic, hierarchical, query-adaptive, proposition-level, late-chunking, and contextual retrieval methods, drawing on ten recent studies spanning 2024–2026. We identify three underexplored research axes: the effect of token-overlap on cross-boundary evidence recovery in clinical text; the comparative adequacy of NLTK punkt versus biomedical sentence segmenters (scispaCy); and the location of the context-density cliff in PubMed prose. These axes map directly to an ongoing empirical study comparing sixteen chunking conditions on approximately one thousand PubMed abstracts across five MeSH-defined disease topics.

Keywords: Retrieval-Augmented Generation, chunking strategies, biomedical NLP, PubMed, semantic chunking, fixed-token chunking, sentence segmentation, context window, FAISS, embedding models.

1. INTRODUCTION

The integration of external knowledge retrieval into generative language systems — commonly described as Retrieval-Augmented Generation — was formalised by Lewis et al. (2020) in a seminal NeurIPS paper that combined a dense passage retriever with a seq2seq generator [1]. Since that foundational work, the RAG paradigm has expanded rapidly into high-stakes verticals including clinical decision support, drug information synthesis, and systematic literature review automation. The practical promise is clear: LLMs that hallucinate when operating from parametric memory alone can be constrained to verified source material if the retrieval component surfaces relevant, faithful context.

However, retrieval quality is not merely a function of the embedding model or the similarity search algorithm. It is fundamentally determined by how the source corpus is partitioned into retrievable units before any query is ever posed. Chunking decisions govern the semantic coherence of individual units, the extent to which inter-sentence relationships are preserved or severed, and the total volume of text that ultimately enters the LLM context window. Too-small chunks risk fragmenting evidence spread across multiple sentences; too-large chunks dilute the retrieval signal and inflate context length. The optimal partition is almost certainly domain- and task-dependent, yet most deployed RAG systems default to fixed-size chunking because it is

simple and deterministic [3].

Biomedical text exacerbates every one of these challenges. A typical PubMed abstract describes a study population, an intervention, an outcome measure, and a statistical result — often within a single sentence that spans sixty words and four named entities. The sentence boundary tokeniser used in the majority of NLP pipelines (NLTK punkt) was trained on general news corpora; it misclassifies the period in 'i.v.', 'p.o.', 'Dr.', and similar biomedical abbreviations as sentence terminals, producing spurious short chunks that destroy clinical meaning [2]. Domain-adapted segmenters such as scispaCy address this limitation but have not been benchmarked head-to-head against their general-domain counterparts in a retrieval-generation evaluation framework.

The motivation for the present survey arises from a specific research gap: as of early 2026, no published study has conducted a systematic, multi-metric comparison of sentence-boundary, semantic, and proposition-level chunking on PubMed biomedical abstracts. The survey in [3] states this explicitly — a comprehensive biomedical chunking evaluation remains absent from the literature. The sections below trace the arc from fixed-size baselines (Section 2) through the emerging paradigms of hierarchical, query-adaptive, and contextual chunking (Sections 3–6), situate the present work within that landscape (Section 7), and present a comparative synthesis (Section 8).

2. FIXED-SIZE AND SENTENCE-BOUNDARY CHUNKING

2.1 Fixed-Token Chunking as the Baseline

Fixed-size chunking partitions a document into non-overlapping (or partially overlapping) windows of a specified token count. Its appeal is computational: it requires no linguistic analysis, processes any language identically, and completes in under one second per document for corpora of several thousand abstracts [3]. The LangChain RecursiveCharacterTextSplitter, released in 2022, popularised the pattern among practitioners and has since become the de facto default in open-source RAG implementations [4].

The primary variable in fixed-size chunking is window size, typically expressed in tokens or words. Common choices in the literature are 128, 256, and 512 tokens [3, 5]. A secondary variable is overlap — the number of tokens shared between adjacent chunks. Overlap is theoretically motivated by the concern that semantically related content may straddle a chunk boundary; a non-zero overlap allows both flanking chunks to carry partial evidence from the boundary region. However, the empirical evidence for this intuition is equivocal. Smigielski et al. (2026) find that fixed-size chunking with window 512 and 50-token overlap is the most computationally robust method across eight datasets, but do not isolate the overlap contribution from the window-size effect [3]. The hypothesis that overlap provides no additional retrieval benefit — derived from the Natural Questions benchmark — may not transfer to biomedical text, where a single drug-disease-dosage relationship routinely spans two or three sentences.

Qu et al. (NAACL 2025) examine whether semantic chunking — which dynamically identifies split points using embedding similarity — justifies its computational overhead compared to the fixed baseline [5]. Their finding is nuanced: semantic chunking improves retrieval F1 on long-form documents with coherent topical sections, but on shorter texts — including abstracts — the improvement shrinks to within noise. This result motivates the inclusion of PubMed abstracts (median length approximately 250 words) as a test domain: they are short enough that semantic boundary detection may not outperform a well-tuned fixed splitter.

2.2 Sentence-Boundary Segmentation

Sentence-boundary chunking treats each detected sentence as an atomic retrieval unit, occasionally merging short sentences to reach a minimum token threshold. The principal advantage over fixed-size splitting is that it never severs a sentence mid-clause, preserving predicate-argument structure. The principal risk is tokeniser error: false boundary detections create degenerate chunks while missed boundaries produce oversized ones.

NLTK's punkt tokeniser implements an unsupervised algorithm that learns abbreviation lists and collocations from training text. When applied to PubMed content, it encounters a large set of biomedical abbreviations — 'i.v.' (intravenous), 'p.o.' (per os), 'vs.' (versus), 'et al.' — that are absent from or underrepresented in its training corpus. The consequence is a systematic over-segmentation at these tokens, producing fragments such as 'Patients received 10 mg metformin i.' and 'v.' as separate chunks — neither of which is retrievable in isolation.

scispaCy, developed by Neumann et al. at the Allen Institute for Artificial Intelligence, provides a sentence segmenter trained on biomedical corpora including MedMentions and BC5CDR [6]. Its dependency parse-based segmentation respects domain abbreviations and handles structured abstract sections (Background, Methods, Results, Conclusions) as distinct paragraph units. No published RAG study has compared NLTK punkt and scispaCy sentence chunking under identical retrieval-generation conditions on PubMed text, leaving open the question of whether the segmenter choice materially affects downstream answer quality.

3. SEMANTIC AND EMBEDDING-BASED CHUNKING

3.1 *Embedding-Similarity Boundary Detection*

Semantic chunking replaces the fixed-window rule with a data-driven criterion: a split is inserted wherever the cosine similarity between the embeddings of adjacent text units falls below a threshold θ . The intuition is that a sharp similarity drop signals a genuine topic transition. The approach was implicit in TextTiling (Hearst, 1997) — which used term-frequency vectors rather than dense embeddings — and has been updated for the dense-vector era by several recent frameworks [7]. The most widely used open-source implementation employs all-MiniLM-L6-v2 (sentence-transformers) at $\theta \approx 0.75$.

Smigielski et al. (2026) evaluate two embedding-based semantic chunking variants — Recursive Semantic and Sequential HAC (hierarchical agglomerative clustering) — across eight benchmark datasets including NaturalQuestions, SQuAD, and Qasper [3]. Recursive Semantic achieves results comparable to the best fixed-size baseline on dense-retrieval benchmarks but at processing times that are two to three orders of magnitude longer. On the GutenQA dataset (literary text), semantic chunking outperforms fixed chunking by 4–6 F1 points; on NaturalQuestions (short factoid questions), the advantage disappears. The authors conclude that chunking complexity should be matched to document complexity — a principle with direct implications for short biomedical abstracts.

An underexplored dimension is the interaction between the chunker's embedding model and the retriever's embedding model. If both use the same representation space, semantic split points align with retrieval-relevant features; if they differ — for example, MiniLM is used to segment while PubMedBERT is used to retrieve — the boundary placements may be suboptimal for the retrieval task. No prior work has examined this cross-model interaction in a biomedical setting.

3.2 *Max-Min Semantic and Graph-Based Variants*

Beyond the pairwise adjacency criterion, Smigielski et al. (2026) also evaluate Max-Min Semantic chunking (which selects split points to maximise the minimum within-chunk similarity) and GraphSeg (which builds a

sentence similarity graph and partitions it into communities) [3]. Both require pairwise similarity computations across all sentence pairs, scaling as $O(n^2)$ in the number of sentences. On the corpora examined, neither outperforms Recursive Semantic consistently, yet both incur substantially higher computational costs. For abstracts of 10–20 sentences, the $O(n^2)$ cost is negligible; but at the document level, these methods become impractical without GPU acceleration.

4. PROPOSITION-LEVEL AND LLM-BASED CHUNKING

4.1 Dense Passage Retrieval at Proposition Granularity

Chen et al. (2024) proposed Dense X Retrieval, in which a source document is decomposed into atomic propositions — minimal, self-contained factual statements — using an LLM as the decomposer [8]. Each proposition is then embedded and indexed independently. At query time, retrieval operates over propositions rather than passages; a post-retrieval step reassembles the original passage context around each matched proposition before passing it to the generator. The key claim is that proposition-level indexing increases retrieval precision by eliminating the noise from co-located but irrelevant sentences.

The cost, however, is substantial. Smigielski et al. (2026) reproduce DenseX (which they label as their 'proposition' condition) and report average processing times exceeding fifteen hours per dataset — versus under one second for fixed-size chunking [3]. On five of eight evaluated datasets, DenseX fails entirely due to API timeouts or memory overflow. On the datasets where it completes, it does not consistently outperform fixed-size chunking. The authors characterise this as evidence that 'chunk quality matters more than chunk quantity' — the granularity advantage of propositions is not sufficient to compensate for the noise introduced by LLM decomposition errors and the increased retrieval surface area.

LumberChunker (Duarte et al., 2024) adopts a related but less granular approach: it uses an LLM to detect semantic shift points in running text and inserts splits at those points rather than decomposing each sentence into sub-propositional units [9]. Smigielski et al. evaluate LumberChunker and report that it also times out on several benchmark datasets [3]. Where it does complete, LumberChunker performs comparably to Recursive Semantic chunking at substantially higher computational cost. Both DenseX and LumberChunker require LLM API calls per document, making them sensitive to rate limits, API pricing, and non-determinism in the chunker LLM — a practical concern for large-scale biomedical corpora.

4.2 Meta-Chunking and Mixture-of-Granularities

Zhao et al. (2025) introduce Meta-Chunking, which operates at two levels: a Margin Sampling criterion assigns each split candidate a boundary probability derived from the perplexity gradient of a small language model, while a Dynamic Combination strategy merges adjacent chunks based on their embedding similarity to the query at run time [10]. The approach is positioned as a middle ground between static proposition-level splitting and fully dynamic query-adaptive methods. On the RULER long-document benchmark, Meta-Chunking outperforms fixed-size chunking by 6.2 points in answer accuracy, though the evaluation is restricted to general-domain corpora.

Zhong et al. (2025) propose a Mixture-of-Granularities (MoG) framework that routes each query to the chunking granularity best matched to the query type [11]. Factoid queries are routed to proposition-level chunks; thematic queries to sentence-level chunks; broad survey queries to paragraph-level chunks. The routing decision is made by a lightweight classifier trained on the query embeddings. MoG achieves consistent improvements over any single granularity in isolation, but adds a routing component that must be trained on domain-specific data — an obstacle for zero-shot biomedical deployment.

5. HIERARCHICAL AND QUERY-ADAPTIVE CHUNKING

5.1 HiChunk: Hierarchical Structure with Auto-Merge Retrieval

Lu et al. (2025) at Tencent Youtu Lab identify a fundamental tension in RAG chunking: small chunks yield high retrieval precision but insufficient context for generation; large chunks provide context but dilute the retrieval signal [12]. They propose HiChunk, a framework that resolves this tension by explicitly modelling document hierarchy. A fine-tuned Qwen3-4B model assigns each segment to one of three hierarchical levels (section, paragraph, clause) based on its structural role in the document. At retrieval time, the Auto-Merge algorithm identifies clusters of matched fine-grained chunks and promotes them to the coarser parent context before passing to the generator.

To evaluate HiChunk fairly, Lu et al. construct HiCBench — a manually annotated benchmark with two tasks: T1 (evidence-dense retrieval, where a single document contains multiple relevant pieces) and T2 (evidence-sparse retrieval, where relevant evidence is scattered across the corpus) [12]. On T1 with Qwen3-8B as generator, HC200+AM achieves evidence recall of 81.03%, compared to 72.1% for fixed-size 200-token chunking and 68.4% for LumberChunker. On T2, the gap narrows considerably — HiChunk adds about 3 points over fixed baseline. This task-dependency is instructive: PubMed abstracts, which are structured syntheses of study findings, are closer to evidence-dense documents than evidence-sparse ones, suggesting HiChunk's hierarchical approach may be well-suited to the biomedical domain.

Processing speed, however, distinguishes the approaches sharply. Fixed-size chunking processes a document in under one second; HiChunk requires approximately 5.75 seconds per document owing to the LLM inference step; LumberChunker processes the same document in several minutes [12]. For a corpus of one thousand PubMed abstracts, these figures translate to roughly sixteen minutes (HiChunk) versus hours (LumberChunker), making HiChunk the only LLM-based chunker with practical throughput on a CPU-only research cluster.

5.2 QASC: Query-Adaptive Semantic Chunking

Rastogi (2025) introduces a conceptually distinct departure from all prior methods: Query-Adaptive Semantic Chunking (QASC) decouples the chunking process from the indexing process entirely [13]. Rather than segmenting at index time and retrieving at query time, QASC integrates the user query into the segmentation decision itself. The pipeline operates in three stages: (i) each candidate sentence in the source document is scored by cosine similarity to the query embedding using all-MiniLM-L6-v2; (ii) seed sentences (those exceeding a similarity threshold) are identified as anchors; (iii) a contextual window of $m = 3$ sentences around each seed is assembled into the retrieved chunk, with chunk scores aggregated by maximum seed similarity.

QASC is evaluated on 100 technical documents with 40 queries spanning four types: factoid, topical, comparative, and multi-hop [13]. QASC achieves $F1 = 0.85$ versus 0.76 for standard semantic chunking and 0.72 for the best fixed-size baseline — improvements of 8–12 percentage points and 18–27 percentage points respectively. Faithfulness reaches 0.87, and inter-annotator agreement on human evaluation reaches Fleiss' $\kappa = 0.74$ –0.80 across evaluator panels. Query latency is 380 ms versus 2,850 ms for agentic RAG pipelines.

The main limitation of QASC for the present research context is practical: because chunking is performed at query time rather than index time, QASC cannot pre-compute and cache a static index. Each query incurs a full document scan. For a corpus of one thousand abstracts and one hundred questions at four k-values, this results in 400,000 individual cosine-similarity scans, which may be computationally prohibitive without

vector acceleration. Additionally, QASC's evaluation does not include biomedical text, leaving open the question of whether the MiniLM similarity threshold is well-calibrated for clinical vocabulary.

6. LATE CHUNKING AND CONTEXTUAL RETRIEVAL

6.1 Late Chunking: Embed First, Segment Later

Günther et al. (2024) propose a reversal of the conventional chunking order: instead of segmenting a document and then embedding each chunk independently, late chunking first passes the entire document through the encoder as a single sequence, extracting per-token or per-sentence representations that incorporate full-document context [14]. Segmentation is then applied to these contextualised representations post-hoc, producing chunks whose embeddings reflect not only their local content but also the document's overall argumentative context.

Merola and Singh (2025) evaluate late chunking against contextual retrieval in a unified experimental framework, testing four embedding models: Stella-V5, Jina-V3, Jina-V2, and BGE-M3 [15]. Their dataset combination includes NFCorpus (biomedical retrieval) and MS MARCO (general-domain question answering). On NFCorpus, late chunking with Jina-V3 achieves $\text{NDCG@5} = 0.309$, narrowly below contextual retrieval at 0.317. Late chunking is more computationally efficient — requiring no LLM context-generation step — but shows inconsistent behaviour across embedding models: it improves over standard chunking with long-context encoders (Jina-V3, Stella-V5) but degrades with shorter-context models (BGE-M3). This model-dependency makes late chunking harder to deploy in settings where the embedding model may change.

6.2 Contextual Retrieval: LLM-Prepended Context Prefixes

Anthropic (2024) introduced Contextual Retrieval as an enhancement to the standard chunking pipeline: before embedding each chunk, a short LLM-generated context description (50–100 tokens) is prepended to the chunk text, situating the chunk within its source document [16]. The hypothesis is that isolated chunks lose the document-level cues needed to disambiguate entity mentions and pronominal references; the prepended context restores that information without increasing the retrieval granularity.

Merola and Singh (2025) combine contextual retrieval with BM25 rank fusion and a cross-encoder reranker, testing multiple embedding models on NFCorpus and MS MARCO [15]. The full pipeline — contextual retrieval + BM25 fusion + reranking — achieves the highest NDCG@5 scores in the study (0.317 with Jina-V3 on NFCorpus). The computational cost, however, is substantial: LLM context-generation on GPU requires approximately 20 GB of device memory, and the contextualization step adds roughly one API call per chunk. On a CPU-only system with a corpus of 50,000 chunks, this step alone may require several hours. Within the scope of an abstract-level PubMed corpus (~15,000 chunks), the overhead is more manageable but still non-trivial.

7. CHUNKING STRATEGIES IN THE BIOMEDICAL DOMAIN

7.1 The Evidence-Density Problem in Medical RAG

Clinical and biomedical text exhibits a property that has received insufficient attention in the chunking literature: evidence density. A PubMed abstract reporting a randomised controlled trial may pack the intervention, comparator, primary endpoint, hazard ratio, confidence interval, and p-value into two consecutive sentences. If fixed-size chunking places a boundary between those sentences, a retrieval query for the efficacy outcome will recover the effect size but not the confidence interval — or vice versa. The chunk is

technically relevant but insufficiently complete for accurate answer generation.

Lu et al. (2025) operationalise this concern through HiCBench Task T1 (evidence-dense queries) and demonstrate that the gap between fixed-size and hierarchical chunking is larger on dense-evidence tasks [12]. Their finding aligns with the prior observation by Shi et al. (2023) that irrelevant context in the LLM input degrades performance — a phenomenon they termed 'lost in the middle' [17]. If dense evidence is spread across multiple retrieved chunks, some of which are marginally relevant, the generator's attention may not span all the needed information, leading to incomplete answers.

A 2025 study published in MDPI Bioengineering (PMC12649634) conducted what is arguably the closest prior work to the present research: a comparison of adaptive versus fixed chunking on medical question answering [18]. The adaptive method — which dynamically adjusts chunk boundaries based on medical entity density — achieved 87% answer accuracy versus 13% for a fixed-size baseline on physician-validated clinical questions. The magnitude of this difference far exceeds what general-domain benchmarks would predict, suggesting that biomedical text poses a fundamentally different retrieval problem than Wikipedia or news corpora.

7.2 Embedding Model Choice in Biomedical Retrieval

Most RAG chunking studies use general-domain embedding models (BERT-base, all-MiniLM-L6-v2, GTE, E5) as both the chunking signal and the retrieval encoder. In the biomedical domain, domain-adapted models such as PubMedBERT (Gu et al., 2021) — trained from scratch on PubMed abstracts and PubMed Central full texts — have consistently outperformed general-domain BERT on downstream biomedical NLP tasks including named entity recognition, relation extraction, and question answering [19]. The NeuML/pubmedbert-base-embeddings variant adapts this encoder for dense retrieval via contrastive fine-tuning on biomedical query-passage pairs, producing 768-dimensional representations that may capture drug–gene–disease relationships more faithfully than the 384-dimensional all-MiniLM-L6-v2 embeddings.

Whether this embedding advantage persists across different chunking strategies is unknown. It is conceivable that the benefit of PubMedBERT over MiniLM diminishes for semantically coherent chunks (where both models agree on relevance) but amplifies for fragmented chunks (where biomedical co-reference is needed to infer relevance). This embedding × chunking interaction has not been examined in any published study.

7.3 The Context-Length Cliff in Biomedical Abstractive QA

A well-documented finding in general-domain RAG is that answer quality reaches a plateau — or even declines — once the retrieved context exceeds a threshold of approximately 2,500 words [20]. The proposed mechanism is that LLM attention diffuses over longer contexts, reducing the probability of attending to any single relevant passage. This 'context cliff' is relevant to the choice of k (the number of retrieved chunks): larger k yields higher retrieval recall but may flood the context with irrelevant material.

Whether this cliff occurs at the same threshold in biomedical text is unclear. PubMed abstracts are significantly denser than Wikipedia paragraphs — a 250-word abstract may carry the information equivalent of a 600-word Wikipedia section. If LLM attention is governed by information density rather than word count, the biomedical context cliff may occur at a lower word threshold than the 2,500 observed in general-domain studies. Alternatively, the structured format of IMRAD-style abstracts (Introduction, Methods, Results, and Discussion) may provide salient structural anchors that help the LLM locate relevant content even in longer contexts, pushing the cliff to a higher threshold.

8. COMPARATIVE ANALYSIS

8.1 Summary of Key Methods

Method	Boundary Criterion	Complexity	Best Domain	Biomedical Validated	Key Ref
Fixed-Token	Token count	$O(1)/\text{doc}$	General	Partially	[3,5]
Sentence-NLTK	Punkt tokeniser	$O(n)$	General	No	[2]
Sentence-scispaCy	Dep. parse (biomedical)	$O(n)$	Biomedical	No	[6]
Semantic ($\cos \theta$)	Embedding similarity	$O(n \cdot d)$	Mixed	No	[3,7]
TextTiling	Term freq. valleys	$O(n^2)$	Long docs	No	[7]
LumberChunker	LLM shift detection	$O(n) \cdot \text{API}$	Long docs	No	[3,9]
DenseX / Proposition	LLM decomposition	$O(n) \cdot \text{API}$	Factoid QA	No	[3,8]
HiChunk	LLM hierarchy + AM	$O(n) \cdot \text{LLM}$	Evidence-dense	No	[12]
QASC	Query-at-segment time	$O(n \cdot q)/\text{query}$	Technical	No	[13]
Late Chunking	Post-hoc segment of doc embedding	$O(n \cdot d)$	Long docs	Partial	[14,15]
Contextual Retrieval	LLM prefix per chunk	$O(n) \cdot \text{API}$	Precision-sensitive	Partial	[15,16]
Adaptive (medical)	Entity density	$O(n)$	Biomedical	Yes	[18]

Table 1: Comparative overview of chunking strategies surveyed.

8.2 Computational Cost vs. Performance Trade-Off

A consistent finding across the surveyed literature is that computational complexity and downstream performance do not scale proportionally. Fixed-size chunking — the least expensive method — demonstrates robust performance across the widest range of datasets in Smigielski et al.'s 2026 benchmark [3]. LumberChunker and DenseX, which rank among the most computationally expensive methods, fail entirely on five of eight datasets due to resource exhaustion. HiChunk achieves the strongest evidence recall on dense-retrieval tasks but requires LLM inference at chunking time, a cost that most research groups can only partially absorb through free-tier API access.

The practical implication for resource-constrained research — including the present study, which runs entirely on CPU without GPU acceleration — is that proposition-level and LLM-based chunkers must be evaluated against their API costs and latency constraints, not merely their retrieval scores. A method that achieves 5 additional F1 points at 300× the compute cost does not represent a Pareto improvement for most deployment scenarios.

8.3 The Research Gap in Biomedical Chunking

The following dimensions of the chunking problem remain either unaddressed or only partially addressed in the biomedical domain:

Research Axis	Status in General-Domain Lit.	Status in Biomedical Lit.
Token overlap in fixed chunking	Partially studied on NQ, SQuAD	Not systematically evaluated on PubMed
NLTK vs. biomedical sentence segmenter	NLTK is de facto standard; no comparison	scispaCy available but not benchmarked in RAG
Semantic vs. sentence chunking	Comparable on short docs (Qu et al., 2025)	Unknown on 250-word PubMed abstracts
Proposition/LLM chunking scalability	Fails at scale (Smigielski et al., 2026)	Not evaluated on biomedical text
Context-length cliff threshold	~2,500 words (Wikipedia corpora)	Unknown for denser biomedical text
Embedding model × chunking interaction	Not studied	Not studied (MiniLM vs. PubMedBERT)
Adaptive/entity-aware chunking	MoG, Meta-Chunking exist	One study (PMC12649634); needs replication
Query-adaptive chunking (QASC)	Evaluated on technical docs	Not evaluated on PubMed

Table 2: Open research questions in biomedical RAG chunking.

9. THE PRESENT STUDY: EXPERIMENTAL FRAMEWORK

9.1 Corpus and Question Set

The empirical work reviewed in this survey motivates an ongoing study that addresses the open questions in Table 2. The corpus consists of approximately one thousand PubMed abstracts collected via the Entrez API, distributed across five MeSH-defined disease topics: hypertension, type 2 diabetes mellitus, lung neoplasms, COVID-19, and Alzheimer's disease. These topics were selected to span chronic disease, infectious disease, oncology, neurology, and metabolic medicine — covering a representative cross-section of the clinical literature. Abstracts shorter than fifty words are excluded to eliminate single-sentence records. One hundred annotated questions (twenty per topic) are paired with gold reference answers and source PMIDs, enabling both retrieval and generation evaluation.

9.2 Chunking Conditions

Sixteen chunking conditions are evaluated in a two-phase design. Phase A (Part A) evaluates twelve fixed-token conditions formed by the Cartesian product of three window sizes (128, 256, 512 words) and four overlap percentages (0%, 10%, 20%, 40%). The overlap manipulation directly tests Hypothesis H1 — whether overlap provides a retrievability benefit in biomedical text. Phase B (Part B) adds four advanced conditions: NLTK sentence chunking, scispaCy sentence chunking, embedding-similarity semantic chunking (threshold $\theta = 0.75$, all-MiniLM-L6-v2), and proposition-level chunking (atomic decomposition via Groq API Llama-3-70B).

9.3 Embedding and Retrieval

Each chunking condition is indexed using two embedding models: all-MiniLM-L6-v2 (384 dimensions, general-domain) and NeuML/pubmedbert-base-embeddings (768 dimensions, biomedical domain). Indexes are built using FAISS IndexFlatIP on L2-normalised vectors, enabling cosine similarity search without GPU. Retrieval is performed at four k-values (1, 3, 5, 10), yielding a full factorial of 16 conditions × 2 embeddings × 4 k-values × 100 questions = 12,800 evaluation runs, plus 800 additional runs for the best fixed-token anchor

in Part B.

9.4 Generation and Evaluation

Retrieved chunks are assembled into a context string and passed to Gemini-2.0-Flash-Lite (temperature = 0.0, maximum 512 output tokens) with a constrained system prompt requiring the model to answer from provided context only. A four-second inter-call sleep respects the free-tier rate limit of fifteen requests per minute. Six metrics are recorded for each run: answer relevance (cosine similarity of query and answer embeddings), ROUGE-L F1 against the gold reference, faithfulness (token-level precision overlap between answer and context), retrieval recall (proportion of gold PMIDs recovered), context word count (the H3 independent variable), and end-to-end latency in milliseconds.

Hypothesis	Variable Tested	Metric	Expected Finding
H1: Overlap benefit	Overlap % (0, 10, 20, 40)	Retrieval recall, ROUGE-L	Overlap improves recall on PubMed due to cross-boundary clinical evidence
H2: NLTK vs. scispaCy vs. semantic	Chunker type (fixed, sent_nltk, sent_scispacy, semantic)	Answer relevance, faithfulness	scispaCy and/or semantic outperform NLTK on biomedical text
H3: Context-length cliff	Context words (varies with k and window size)	Answer relevance at increasing context word count	Cliff at a different threshold than general-domain 2,500 words

Table 3: Research hypotheses and their operationalisation.

9.5 Novelty and Contributions

Relative to the prior work surveyed in Sections 2–7, the present study makes the following contributions. First, it provides the first head-to-head comparison of NLTK punkt and scispaCy sentence chunking within a retrieval-generation evaluation framework on PubMed text — addressing a gap explicitly noted by Smigielski et al. (2026) [3]. Second, it introduces PubMedBERT as a second embedding model alongside MiniLM, enabling direct measurement of the embedding × chunking interaction that has not been studied in any prior chunking benchmark. Third, it tests proposition-level chunking on biomedical abstracts — a domain where atomic facts are both more precise (drug-disease-dosage triplets) and more fragile under LLM decomposition (complex statistical reporting) than in general-domain text. Fourth, the context-cliff hypothesis (H3) is operationalised at the abstract level, where the density of clinical information per word differs markedly from Wikipedia or news corpora.

10. DISCUSSION

The trajectory of chunking research from 2020 to 2026 reveals a familiar pattern: simple baselines established early, more complex alternatives proposed and partially validated, and the field now entering a phase of critical reassessment. The reassessment is driven by two convergent findings: that LLM-based chunking methods (DenseX, LumberChunker) fail at scale despite theoretical appeal [3], and that the domain-specificity of chunking decisions has been systematically underweighted in benchmark design [18].

For the biomedical domain specifically, the literature points toward a productive middle ground: methods that are smarter than fixed-size splitting but more tractable than full LLM decomposition. Semantic chunking satisfies this criterion computationally, though its performance advantage on short abstracts remains to be demonstrated [5]. scispaCy sentence chunking may provide the biomedical adaptation needed without the API

cost of LLM-based methods [6]. HiChunk's Auto-Merge retrieval is conceptually appealing for evidence-dense biomedical documents but requires a fine-tuned model that may not generalise to the abstract-only setting [12].

The overlap hypothesis (H1) deserves particular attention. The clinical literature contains many instances where a drug is named in one sentence and its side effect reported in the next — a cross-boundary evidence pattern that overlap is designed to address. Whether overlap improves retrieval in this setting by a statistically significant margin is an empirical question that the present study is positioned to answer for the first time.

The context-cliff hypothesis (H3) has practical relevance beyond academic benchmarking. If the cliff occurs at 1,500 words in PubMed text (rather than 2,500 in Wikipedia), then deploying $k = 10$ with 256-word chunks would routinely exceed the optimal context length. Clinicians using RAG-based decision-support tools would receive answers generated from over-loaded contexts, with attendant risks of information omission or attention diffusion. Identifying the correct cliff threshold is therefore a safety-relevant finding, not merely a performance metric.

11. CONCLUSION

This survey has reviewed the principal chunking strategies proposed for Retrieval-Augmented Generation, tracing their development from fixed-size windowing through sentence-boundary segmentation, semantic boundary detection, hierarchical and LLM-based decomposition, query-adaptive segmentation, late chunking, and contextual retrieval. Ten studies from 2024–2026 have been examined in detail, and their findings synthesised with respect to the biomedical domain.

The central conclusion is that the biomedical domain remains significantly under-represented in chunking research: only one published study (PMC12649634) has evaluated chunking strategies specifically on medical text, and none has conducted a multi-method, multi-metric comparison on PubMed abstracts covering diverse disease topics. The present study addresses this gap through a sixteen-condition, two-embedding, full-factorial experiment on one thousand PubMed abstracts, with hypotheses specifically designed to surface the mechanisms through which chunking choices interact with biomedical text properties.

The findings — when complete — will provide practitioners with actionable guidance on chunking strategy selection for clinical RAG applications, and will contribute to the broader literature by grounding general-domain findings in a rigorous biomedical evaluation. The dataset, chunking code, and evaluation framework are designed to be fully reproducible on CPU-only hardware, lowering the barrier for future biomedical RAG research.

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